

HTTP/2 DoS Attacks in 5G Networks: Impact Analysis and Anomaly Detection

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Abstract—Fifth Generation (5G) and beyond networks rely on the HTTP/2 protocol for signaling between core Network Functions (NFs). While HTTP/2 vulnerabilities have been exploited to perform various types of Denial of Service (DoS) attacks in web environments, their impact on telecommunication networks remains under-studied. Though secure by design, the 5G Service-Based Architecture (SBA) can be vulnerable to misconfigurations and virtualization exploits, particularly with Mobile Network Operators (MNOs) using hyper-scale technologies. This work addresses the lack of practical studies and analyses on the impact of HTTP/2 attacks on 5G networks, especially given the absence of a 5G-compliant dataset for anomaly detection. Utilizing the open-source free5GC testbed and UERANSIM emulator, we emulate six different HTTP/2 attacks on various NFs within the 5G SBA. We analyze their impact on the network and demonstrate that many of them cause cascading effects on other NFs involved in related jeopardized 5G procedures. Our emulations include both malicious and normal network behavior, resulting in the first 5G anomaly detection dataset that we are aware of. Using CICFlowmeter, we extract flow-based features known for their anomaly detection capabilities and train multiple machine learning models. These models can serve as benchmarks for detecting HTTP/2 attacks in 5G networks.

Index Terms—5G networks, 5G security, HTTP/2, HTTP/2 attacks, 5G SBA security, 5G dataset, machine learning, anomaly detection.

I. INTRODUCTION

THE Fifth Generation (5G) mobile networks introduce remarkable advancement in communication technology [1]. Designed to accommodate a wide range of applications, 5G networks enhance everything from broadband mobile services to the Internet of Things (IoT) and critical communication tasks [2]. At the heart of 5G architecture is the innovative Service-Based Architecture (SBA) (Fig. 1), which coordinates specialized Network Functions (NFs) to manage network operations [3]. These NFs utilize standardized protocols for their efficient communication. The Third Generation Partnership Project (3GPP) adopted the Hypertext Transfer Protocol Version 2 (HTTP/2) as their

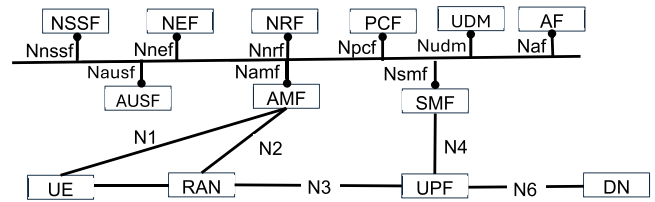


Fig. 1. 5G service-based architecture [3].

communication protocol in 5G SBA [4], [5]. 5G employs a Service-based Interface (SBI) (i.e., nnef, namf, nsmf, etc.) using an Application Programming Interface (API) over HTTP/2 [4] in the Control Plane (Fig. 1).

HTTP/2 introduces significant improvements over its predecessor, HTTP/1.1, providing features like stream multiplexing, header compression, and optimized connection management [6]. These enhancements are crucial for handling the increased signaling traffic and dynamic interactions between 5G NFs. HTTP/2, a fundamental component of the 5G SBA, enables reduced latency, efficient resource allocation, and scalable operations among NFs [4]. By harnessing the protocol features, HTTP/2 ensures robust and seamless communication [4].

To ensure a resilient and dependable experience for 5G users, SBA includes crucial functionalities such as NF registration, discovery, and authorization, along with security measures for the SBI [7]. These functions are supported by the standardized HTTP/2 protocol, secured with Transport Layer Security (TLS), enhancing both security and the system's agility and scalability. Nonetheless, securing 5G SBA presents notable challenges. For instance, NFs can enable on-demand firewall and Intrusion Detection Systems (IDS) to block or reroute malicious traffic. However, the design of such dynamic NF-based security mechanisms is yet to be explored [8]. It is imperative to investigate emerging security challenges specific to the 5G SBA and its associated technologies and protocols. There is a significant need for more research into the cybersecurity impacts of web-based technologies such as HTTP/2 on SBA.

Although 5G network security has been the focus of extensive research, most studies concentrate on well-known threats such as jamming, Denial of Service (DoS), eavesdropping, and Man-in-the-Middle (MITM) attacks [2]. Other works [1], [9], [10] explore the implications of virtualization in 5G SBA, revealing how the use of containers, VNFs, and cloud-native infrastructure broadens the attack surface. However, far fewer studies have critically examined the security implications of

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adopting HTTP/2 as the core signaling protocol in 5G SBA. Only a limited number of efforts [11], [12], [13] have addressed this emerging threat landscape, usually focused on signaling vulnerabilities or specific features of HTTP/2. In particular, the work in [13] investigates HTTP/2 Stream Multiplexing Attacks (SMAs) in a 5G environment but does not release its dataset or extend the analysis to other classes of attacks. Similarly, [14] introduces fuzzing and replay methodologies using 5Greplay, but does not simulate these threats in a full 5G testbed, thus lacking empirical information on their operational impact.

These limitations highlight a critical gap in the literature related to the absence of a publicly available dataset, or a comprehensive experimental analysis of HTTP/2 attacks affecting 5G NFs. Furthermore, existing anomaly detection efforts have not focused on HTTP/2 threats within a standalone 5G SBA environment. To bridge this gap, our work makes several novel contributions motivated by the shortcomings of prior studies [11], [12], [13], [14], [15]:

- We leverage the open-source free5GC [16] core and UERANSIM [17] UE/RAN emulator, aligned with 3GPP standards [18]. We emulate realistic 5G control plane operations and inject six distinct HTTP/2-based attacks, including variations of stream multiplexing, rapid reset, and slow rate attacks. Unlike previous studies that focus on isolated NFs or simulated HTTP/2 traffic, our emulations capture the full impact on NF-to-NF signaling, cascading failures, and degradation of service availability across the entire SBA.
- We build a novel dataset that includes both benign and malicious traffic flows generated during the emulation of realistic 5G procedures and attacks. To the best of our knowledge, this dataset is the first to capture HTTP/2 behavior within a fully functional 5G SBA environment. It is designed to support advanced research in areas such as adversarial robustness, transfer learning, and adaptive anomaly detection. The dataset will be made publicly available upon acceptance.
- We pre-process our 5G HTTP/2 dataset to extract flow-based features that are widely known in the literature for their ability to distinguish between normal and malicious behaviors. We evaluate three unsupervised machine learning models for detecting HTTP/2-based anomalies: Autoencoder (AE), LSTM-Autoencoder, and Isolation Forest (IF). The AE model builds on our prior work [13] which focused on detecting HTTP/2 SMAs. In this work, we test its ability to generalize across all six attack types. LSTM-AE, inspired by its successful use in detecting non-HTTP/2 attacks in 5G SBA [19], demonstrates the advantage of modeling temporal dependencies. Our evaluation shows that LSTM-AE achieves the highest average F1-score of 92.24%, confirming the viability of using flow-based models for HTTP/2 attack detection in real 5G environments.

The paper is organized as follows: Section II gives a background on the 5G networks, then an overview of related works is presented in Section III. The threat model for HTTP/2 attacks and its variants are explored in Section IV. Furthermore, Section V outlines the experimental setup, including the normal

and malicious network emulations. In addition, we conduct a thorough analysis of the impact of HTTP/2 attacks on 5G networks with protection measures in Section VI. Section VII presents our dataset. Our model results are presented in Section VIII. The paper concludes in Section X.

II. BACKGROUND ON HTTP/2 IN 5G

The HTTP/2 protocol was developed to address the shortcomings and performance limitations of its predecessor, HTTP/1.1 [6]. HTTP/2 uses *stream* for *Request/Response* exchanges, where each message is either a request or a response. HTTP/2 messages are built from HTTP/2 frames. An HTTP/2 *frame* is the smallest unit of communication within an HTTP/2 connection, encoded in binary. HTTP/2 frames include (1) *HEADERS* frame for opening streams and contain different header fields in the form of key-value pairs, (2) *DATA* frame carries a payload, and (3) *SETTINGS* frame carries configuration parameters that impact the communication [6]. For example, *SETTINGS_MAX_CONCURRENT_STREAMS SETTINGS* allows an HTTP/2 client/server to limit the maximum number of concurrent streams over a single HTTP/2 connection with its peers and enables the HTTP/2 stream multiplexing features. Other HTTP/2 features include flow control, and header compression, among others [6]. For instance, header compression minimizes the redundancy and size of header data, which minimizes bandwidth utilization [6].

3GPP adopted the HTTP/2 protocol in 5G to facilitate signaling between 5G NFs given its secure, reliable, and bidirectional communication [2]. These benefits position HTTP/2 as a proficient choice for 5G NFs communication, enabling robust, efficient, and scalable network operations. This communication takes the form of *Request/Response* or *Subscribe/Notify*, enabling seamless interaction between the NF Service consumer (NFC) and the NF Service producer (NFP) [4]. *Request/Response* is used when the NFC requests a service and the NFP responds. In contrast, *Subscribe/Notify* is employed when the NFC subscribes to an NFP event (e.g., Session Management Function (SMF) subscribes to location report to get notified of the last known location of a UE or group of UEs by the Access and Mobility Management Function (AMF) [4], etc.), that causes the NFC to be called back when the event occurs. HTTP/2 enhances efficiency and speed, however, it opens up new avenues for exploitation by malicious actors. For instance, despite 5G SBA being secure by design, it remains susceptible to virtualization exploits and misconfigurations which can be used as entry points to launch attacks exploiting HTTP/2 vulnerabilities [10], [20], [21].

III. RELATED WORKS

A. HTTP/2 Anomaly Detection

Numerous studies examine threats exploiting the HTTP/2 protocol in web environments, such as HTTP/2 stream multiplexing attacks [13], [22], HTTP/2 rapid reset attacks [23], [24], and HTTP/2 slow rate attacks [25], [26]. These studies often suggest anomaly detection methods that are generally less

effective against HTTP/2 threats targeting web environments. For example, [22] shows that the HTTP/2 stream multiplexing feature can be exploited to launch multiple streams over the same connection, overwhelming the server or causing DoS. In HTTP/2 slow rate DoS attacks, attackers send multiple specially crafted incomplete requests that occupy the server's connection queue space, preventing it from processing other requests [25]. They can still exhaust server resources, leading to performance degradation and DoS. To detect such attacks, [25] proposes using a Chi-square test to identify abnormal intervals of HTTP/2 traffic. However, its effectiveness varies with the attack rate and the chosen detection interval. The same authors later developed an event sequence analysis method, achieving high accuracy with minimal computational demands [26], only for this attack. Another method by [27] focuses on identifying HTTP/2 multiplexed asymmetric DDoS attacks by contrasting the behavior of legitimate users and attackers. While effective for DDoS attacks, this approach fails to detect HTTP/2 slow-rate DoS attacks due to the minimal computational overhead and pattern mimicking by attackers. Both slow rate and stream multiplexing attacks exploit various HTTP/2 parameters, leading to a DoS. One of the latest attacks related to HTTP/2 is the HTTP/2 rapid reset attack, identified as CVE-2023-44487 [24]. This attack exploits the stream multiplexing feature by resetting streams currently handling requests using the *RST_STREAM* frame. The mitigation for this attack involved bounding the number of simultaneously executing handler routines to a defined limit (`SETTINGS_MAX_CONCURRENT_STREAMS=200`), preventing server overload by queuing incoming requests until a current request is completed. If the queue becomes excessively long, the server terminates the connection.

With the adoption of the HTTP/2 protocol by 3GPP as a signaling protocol in 5G networks, it becomes crucial to understand its security implications. To date, to the best of our knowledge, [13] remains the only study that has emulated HTTP/2-related attacks in a 5G SBA environment. This research focused on anomaly detection systems that monitor stream patterns to identify malicious behaviors, specifically targeting only HTTP/2 stream multiplexing attacks within the context of a 5G network. Another work [14], proposed testing methodologies using an open-source solution called 5Greplay, allowing network operators to defend against flooding and fuzzing attacks. However, the authors did not focus on the impact of these attacks on 5G networks, as their study did not employ a 5G testbed. Additionally, the HTTP/2 dataset in their research was created using MMT-DPI, a tool developed to parse and mutate HTTP/2 packets. Unlike [13], [14], the current paper significantly expands their scopes by targeting multiple HTTP/2-related attack types, including slow-rate and rapid-reset attacks, in addition to stream multiplexing within 5G SBA. Furthermore, our dataset is based on a 5GC testbed that follows 3GPP standardization.

B. HTTP/2 Datasets

Several prior studies have leveraged publicly available datasets such as CICIDS2018, CICDDoS2019, and datasets derived from 4G-LTE and traditional HTTP/1.1-to-HTTP/2

transitions [27], [28], [29], [30], [31]. These datasets have proven useful for general network anomaly detection and have demonstrated the value of flow-based features in identifying threats such as DoS attacks, unauthorized access, and traffic manipulation. Flow features such as duration, volume, and packet counts provide a compact yet powerful representation of network behavior, especially useful in environments with limited visibility into payload data. However, these datasets fall short when applied to 5G SBA scenarios. First, most of them were created in pre-5G contexts and do not reflect the architectural changes, service-based interactions, or traffic characteristics of standalone 5G networks. Second, while a few recent efforts have attempted to model HTTP/2 specific behavior and attacks, such as those using stream multiplexing in 5G [13], they either rely on private datasets or simulate traffic without grounding in a real 5G SBA testbed. Furthermore, works that use 5G testbeds often focus on non-HTTP/2 protocols and do not capture the application layer characteristics unique to HTTP/2-based service communication in 5G [32], [33], [34].

This lack of a public, domain-specific dataset limits researchers' ability to design, evaluate, and compare anomaly detection methods tailored for real-world 5G HTTP/2 traffic. The need for such datasets is especially pressing given the growing adoption of HTTP/2 in 5G core networks and the increasing sophistication of potential attacks. In response to this gap, our work contributes a novel dataset that captures both normal and malicious traffic in a controlled 5G SBA environment using real NF interactions and HTTP/2 communication. By emulating diverse HTTP/2-specific attacks and collecting flow-level data in a fully standalone 5G testbed, we provide the community with a resource that better reflects the operational realities of modern 5G deployments. This significantly advances the field of 5G security, facilitating further exploration and development of effective anomaly detection and security measures for safeguarding 5G networks.

IV. HTTP/2 ATTACKS IN 5G NETWORKS

Although secure by design, the 5G SBA can still experience some attacks resulting from virtualization exploits, misconfigurations, and its HTTP/2 signaling protocol vulnerabilities. In the following, we shed light on some HTTP/2 attacks in 5G networks while detailing the vulnerabilities they exploit and their related threat models.

A. Assumptions

HTTP/2 attacks in 5G networks can be performed through misconfigured or compromised NFs. We consider the following assumptions for HTTP/2 attacks, assuming that attackers compromise the NFc and use it to attack the NFp or vice versa.

- 1) *Attacker compromises an NFc*: Many standardization documents discuss threats brought by NFV and virtualization technologies (e.g., container, virtual machines, etc.) to telecommunication networks and 5G [20]. The adoption of hyper-scale cloud by mobile operators extends the attack surface of their network and makes their NFs vulnerable [20]. An attacker can compromise 5G NFs deployed

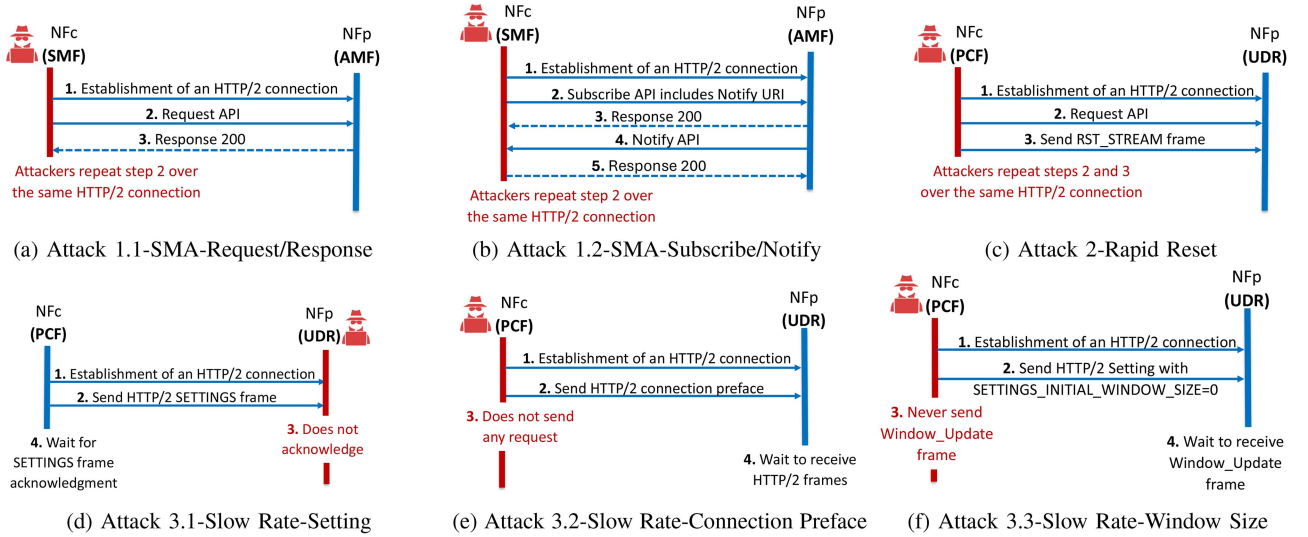


Fig. 2. HTTP/2 attacks in 5G SBA.

on docker containers in the cloud, by exploiting docker vulnerabilities to perform container escape (i.e., CVE-2016-5195 [35], CVE-2019-5736 [36], and CVE-2023-20864 [37]) [38]. Breach of isolation between network slices served by the same compromised NF can also be exploited by attackers [39], [40].

- 2) *NFc can successfully authenticate with the NFp*: We assume that if TLS is used, the malicious NFc can still authenticate with the NFp as the attacker has access to its public/private key pairs.
- 3) *NFc is authorized to access NFp services*: We assume that the malicious NFc has already acquired OAuth2.0 access tokens to the NFp services. These tokens are cached and can be reused by the attacker [2], [41]. Alternatively, the malicious NFc can request new access tokens from the NRF given that it can successfully authenticate with it (i.e., assumption 2). An attacker can exploit vulnerabilities in network slicing and service authorization, as noted in [40], to access NFp services.
- 4) *Attacker has access to UE information*: As some network services require exchanging UE information (e.g., Subscription Permanent Identifier (SUPI)) [42], we assume that the attacker can gain access to such information by monitoring NFc communications or even by requesting such information from other NFs.

We should note that these attacks are not new or novel. HTTP/2 attacks are general attacks that can occur in a web environment as they exploit HTTP/2 features. However, their variations, applicability, and impact on a 5G network are different from a web environment due to the specificity of the 5G SBA and the 5G procedures. In 5G, any NF can act as a service consumer (i.e., HTTP/2 client) or producer (i.e., HTTP/2 server), enabling attacks from both HTTP/2 clients and servers. However, in a web environment, attacks are generally initiated from HTTP/2 clients to the web server. Although exploited in the web, applying these attacks in a 5G environment requires

leveraging 5G-specific APIs, making these attacks more challenging to perform than in a web, where APIs are not necessarily used. Furthermore, their impact on a 5G network can be more disruptive than in a web environment, given the dependencies and interactions existing between the different 5G NFs, as we show in Section VI.

B. Attack 1: HTTP/2 Stream Multiplexing Attack (SMA)

To perform an HTTP/2 SMA, attackers send multiple requests, as much as the NFp allows in the HTTP/2 SETTINGS_MAX_CONCURRENT_STREAMS, into a single HTTP/2 connection. By default, the NFc can send up to 2,147,483,647 (default value of SETTINGS_MAX_CONCURRENT_STREAMS) streams per HTTP/2 connection [6]. Attackers can trigger HTTP/2 SMA in two ways within 5G SBA, either by employing the *Request/Response* or the *Subscribe/Notify*.

1) *Attack 1.1-SMA-Request/Response*: In an HTTP/2 SMA using *Request/Response* (Fig. 2(a)), attackers compromise NFc and send multiple requests over a single HTTP/2 connection towards NFp. Attackers repeat this behavior over multiple HTTP/2 connections which results in a DoS on the NFp.

2) *Attack 1.2-SMA-Subscribe/Notify*: According to 3GPP [4], the *Subscribe/Notify* service operations in HTTP/2 involve two HTTP/2 connections, each handling one direction of traffic. NFc acts as an HTTP/2 client when subscribing to notifications, while NFp functions as an HTTP/2 server. Conversely, the roles are reversed when NFp sends notifications to NFc. As depicted in Fig. 2(b), a compromised NFc establishes an HTTP/2 connection with NFp and sends a subscription request containing a notify URI to signal to the NFp to notify it when the occurrence of the API-related event (e.g., N1N2TransferFailureNotification [43] is triggered). Attackers exploit the event conditions (i.e., UE state is DISCONNECTED [43]) to initiate the notification. Attackers repeat the request with the notification URI to cause an SMA and overwhelm both NFp and NFc. The NFc will be receiving an

excessive number of notifications causing a DoS, while the NFp struggles with the high number of requests and from managing and forwarding the notifications to the NFc, eventually leading to resource exhaustion and DoS on the NFp.

C. Attack 2: HTTP/2 Rapid Reset Attack

Attack 2-Rapid Reset, identified as CVE-2023-44487 [24], exploits the stream multiplexing feature of HTTP/2. It employs the *RST_STREAM* frame to terminate streams that are currently processing requests [6]. In this case, the number of streams that were reset by the *RST_STREAM* frame do not count towards *SETTINGS_MAX_CONCURRENT_STREAMS*. The mitigation for this attack considers counting any request reaching the server, even if it is a *RST_STREAM* frame, as part of the defined maximum stream limit. It involves limiting the number of simultaneously executing handler routines (*SETTINGS_MAX_CONCURRENT_STREAMS* = 200) and prevents server overload by queuing incoming requests until a current request is completed. If the queue becomes excessively long, the server terminates the connection as a safeguard. However, increasing the *SETTINGS_MAX_CONCURRENT_STREAMS* slightly could significantly impact network performance.

In this attack (Fig. 2(c)), attackers compromise the NFc and establish an HTTP/2 connection with the NFp. In this work, we assume that the reset attack is patched, however, we assume that the NFp is misconfigured to allow an unusually high number of concurrent streams (e.g., *SETTINGS_MAX_CONCURRENT_STREAMS* = 1000 instead of the default 200 set in the golang library). The malicious NFc then rapidly generates requests and immediately issues *RST_STREAM* frames for each request across multiple HTTP/2 connections, forcing the NFp to terminate the requests. This flood of reset stream requests can lead to resource exhaustion at the NFp.

D. Attack 3: HTTP/2 Slow Rate Attacks

Another type of HTTP/2 attack is an HTTP/2 slow rate DoS attack which involves attackers sending HTTP/2 frames at a deliberately slow rate to exhaust NFp resources [21], [25], [26]. HTTP/2 slow rate attacks require low bandwidth and are difficult to detect. Attackers exploit the HTTP/2 frame between the NFc and NFp, such as the exchange of *SETTINGS* frame, capitalizing on the design of NFp which waits for certain responses. In this work, we target three variations of HTTP/2 slow rate attacks.

1) *Attack 3.1-Slow Rate-Setting*: HTTP/2 slow Rate-Setting is a slow rate attack that is based on un-acknowledging a *SETTINGS* frame. In a normal HTTP/2 communication scenario, both endpoints must exchange *SETTINGS* frames at the start of a connection and may send them at any other time during the connection. *SETTINGS* frame allows each endpoint to acknowledge the parameters of the connection. When an endpoint receives a *SETTINGS* frame, it should send an acknowledgment response that tells the sender that the *SETTINGS* frame was received and processed. Thus, the slow Rate-Setting attack mainly takes advantage of the *SETTINGS* frame to let the endpoint wait. As depicted in Fig. 2(d), attackers compromise the NFp that has already been authenticated and authorized to access

NFc services. The NFc initiates the first HTTP/2 connection with the compromised NFp, followed by sending a *SETTINGS* frame. However, the malicious NFp does not acknowledge the received *SETTINGS* frame. NFc continues to send numerous requests of *SETTINGS* frame to NFp. Since the malicious NFp consistently fails to acknowledge the HTTP/2 *SETTINGS* frame for all messages received, it can exhaust the available connection pool. This not only blocks other NFs from communicating with the victim NFc but also keeps the connection from NFc open for an extended period.

2) *Attack 3.2-Slow Rate-Connection Preface*: The connection preface is sent from the NFc to inform the NFp that HTTP/2 will be used for further communications. In this attack (Fig. 2(e)), after establishing an HTTP/2 connection, a compromised NFc sends the connection preface to the NFp, prompting it to wait for a GET/POST HTTP/2 request. However, the malicious NFc intentionally withholds any HTTP/2 requests, forcing the NFp to wait until the NFp drops the connection, thus wasting its resources and denying its service to other NFs.

3) *Attack 3.3-Slow Rate-Window Size*: In a standard HTTP/2 connection, both endpoints are required to send an HTTP/2 payload that includes a *SETTINGS* frame with the *SETTINGS_INITIAL_WINDOW_SIZE* field, along with a complete GET request. The *SETTINGS_INITIAL_WINDOW_SIZE* field specifies the sender's capacity to receive data in bytes from its peer. Upon receiving this, the NFp expects that the NFc can receive data of the indicated size. However, attackers exploit this mechanism for a slow rate attack by compromising the NFc (Fig. 2(f)). After establishing the HTTP/2 connection, the malicious NFc sends a *SETTINGS* frame with the *SETTINGS_INITIAL_WINDOW_SIZE* set to zero, falsely indicating no available window space for data reception. The NFp, in turn, holds the data until it receives a *WINDOW_UPDATE* frame that increases the window size. Nonetheless, the malicious NFc intentionally never sends a *WINDOW_UPDATE*, thus causing the NFp to wait till the connection pool is full, resulting in dropping the connection. Thus, this attack exhausts the available connection pool at the NFp and blocks its service for other legitimate UEs.

V. ENVIRONMENT SETUP

In this section, we present the environment that we use to emulate normal and malicious network traffic.

A. Emulation Setup - 5G Testbed

We emulate the HTTP/2 attacks using free5GC [16], an open-source 5G core network testbed, and UERANSIM [17], a UE/RAN emulator, adhering to the 3GPP standard [18]. We [44] as depicted in Fig. 3. The VMs are equipped with Ubuntu 20.04-Focal, 8 virtual Central Processing Units (CPU), and 64 GB of RAM. We use free5GC docker-compose version 3.4.0 [45], which runs NFs in separate containers on the same VM. UERANSIM facilitates the emulation of network traffic and 5G operations, including UE registration, deregistration, and service requests [42], providing crucial insights into network behavior and system performance.

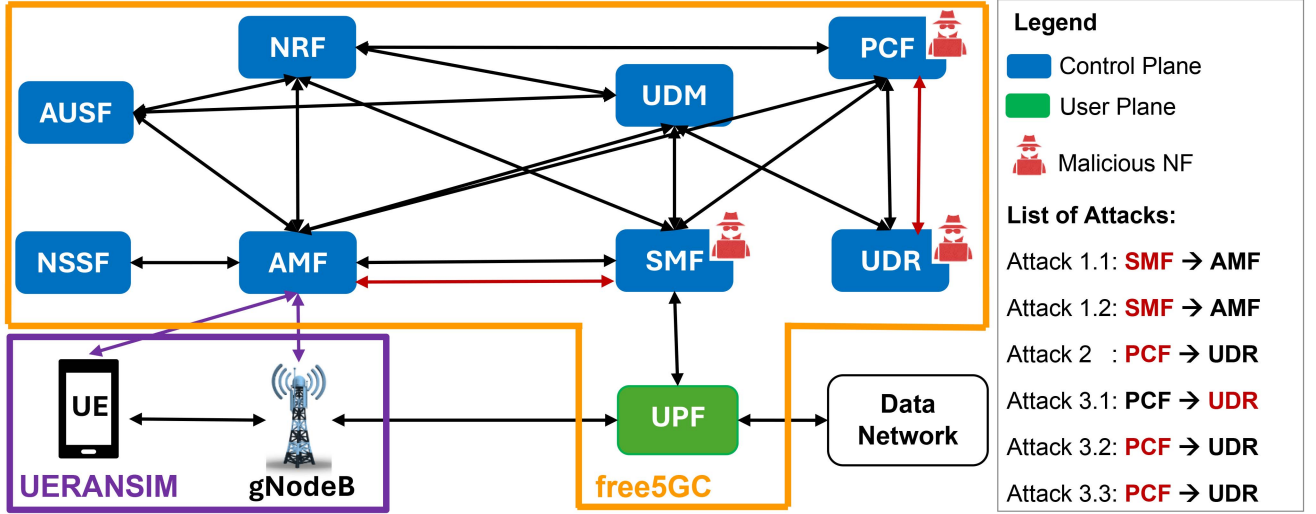


Fig. 3. 5G Testbed with normal and malicious network behaviors.

TABLE I
LOGICAL DEPENDENCY BETWEEN 5G PROCEDURES

Triggered procedure	Possible subsequent procedures
Registration	Uplink, Downlink, UE release PDU session, gNodeB release PDU session, Deregistration
Uplink	Downlink, UE release PDU session, gNodeB release PDU session, Deregistration
Downlink	Uplink, UE release PDU session, gNodeB release PDU session, Deregistration
UE release PDU session	Downlink, Uplink, gNodeB release PDU session, Deregistration
gNodeB release PDU session	Uplink, Downlink, Deregistration
Deregistration	Registration

B. Emulation Configuration

Given the lack of publicly accessible datasets for anomaly detection in the 5G SBA, we employ our emulated 5G testbed to emulate normal and malicious network behaviors. For normal network behavior, we replicate the standard activities of UEs within our 5G testbed by leveraging different 5G procedures implemented in UERANSIM (Table I). These same procedures are also employed to model HTTP/2 attacks and generate malicious traffic.

C. Normal Network Behavior Emulation

To emulate normal network behavior, we consider the arrival of 100 UEs to the network using a Poisson process [46], [47] over two hours. The Poisson process is widely recognized as an effective method for modeling arrival times of events in network traffic due to its ability to capture the randomness of user behavior and service requests over time. For our implementation, we defined the load (number of requests) per 10-minute intervals as [1, 2, 3, 5, 6, 7, 8, 9, 7, 5, 3, 0.5], to reflect the dynamic nature of 5G network traffic. We use 100 UEs in each 10-minute interval, representing one predefined network load. For each load value, we emulate a number of 5G procedures calculated based on the Poisson process for each of these 100 UEs. This approach

follows the principles outlined in [48], where Poisson processes are used to model the arrival of UE requests in realistic network scenarios. Each UE engages in one or more 5G procedures selected from a set provided by UERANSIM (Table I). To ensure that our emulation of 5G normal network behavior is realistic, we follow the 3GPP standard definition of the different 5G procedures and their logical dependencies [18], [42]. We limit these procedures to those available in UERANSIM and which we can emulate. Table I defines the possible subsequent procedure for each triggered procedure by the UE following the 3GPP standard. Given that 5G procedures have logical dependencies and specific order requirements defined by the 3GPP standard, in our emulation, we choose a subsequent procedure ($p + 1$) that follows a procedure p for a UE by randomly selecting it from a predefined list of procedures that logically follow p . The list of the possible subsequent procedures for each 5G operation in UERANSIM, is outlined in Table I. For instance, a UE cannot proceed to deregistration if it has not completed its registration. Moreover, each 5G procedure triggers various communication between NFs. These communications may vary based on the UE state (e.g., CONNECTED, IDLE, DISCONNECTED) and other factors such as network conditions and RAN resources [42].

In our emulation, each UE starts by first registering to the network and then selects a subsequent procedure as detailed in Table I. Hence, the following procedure is randomly chosen from the appropriate options, which include Uplink, Downlink, UE release PDU session, gNodeB release PDU session, and Deregistration procedures. After the registration, let us assume, for example, that the gNodeB release PDU session procedure was selected by the UE. Following its execution, either Uplink, Downlink, or Deregistration procedures can be initiated. Note that these procedures are triggered for the same UE at different times to replicate 5G communications and can switch the UE between different states. For example, (1) UE registers to the network; after a certain period of time, (2) RAN releases the PDU resources allocated to the UE, switching its state to IDLE; (3) Subsequently, a Downlink procedure is triggered from the

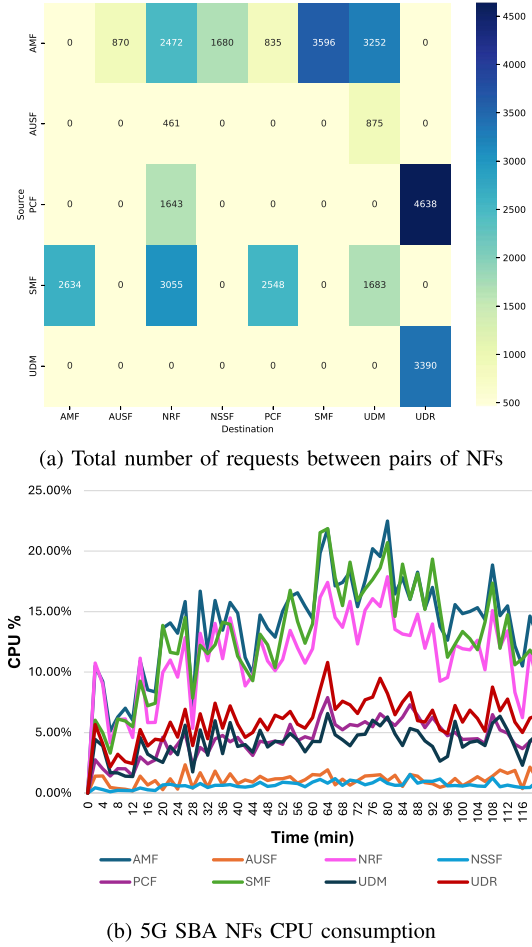


Fig. 4. Benign network traffic.

network to signal to the UE which is in IDLE state, hence, switching its state to CONNECTED.

Fig. 4 highlights the interactions between pairs of NFs observed within our 5G testbed (i.e., control plane). We extract the total number of messages reflecting these interactions between pairs of NFs during 20 minutes of emulations of different 5G procedures and present them in Fig. 4(a). The latter shows that interactions involving the AMF, SMF, UDR, and PCF are more frequent, reflecting the intensive activity associated with Uplink, Downlink, and UDR management procedures during our normal network traffic. This data is crucial as it represents the peak demands for each interaction, offering insights into network load during typical operations.

Along with observing NFs interactions, we monitor resource utilization during normal network traffic by tracking CPU consumption across various NFs over two hours, as shown in Fig. 4(b). We observe that although the CPU load of the different NFs remains under 25%, AMF, SMF, and UDR exhibit higher CPU consumption than other NFs which can be explained by the high number of requests they manage (Fig. 4(a)).

D. Malicious Network Behavior Emulation

5G testbed emulates various HTTP/2 attacks to expose potential vulnerabilities within the 5G SBA. Specifically, we target

procedures such as Uplink, Downlink, UE release PDU session, and UDR Management, frequently used in our 5G testbed. We emulate HTTP/2 attacks (Section IV) where attackers compromise NFc/NFp which are highlighted in red, such as a PCF, UDR, or SMF, as shown in Fig. 3, in addition to UE information (i.e., SUPI). We assume that attackers exploit 30 legitimate UEs out of 100. Through the compromised NFc, each attack is launched using the IMSI of the 30 legitimate UEs, where multiple HTTP/2 connections are established toward the NFp. These connections are configured with a default `SETTINGS_MAX_CONCURRENT_STREAMS=200` in our 5G testbed. Thus, in our attack emulations, the network operates normally for the first 60 minutes, after which the attack is initiated when the load in the network is designed to be around its peak ($load = 8$).

1) *Attack 1: HTTP/2 Stream Multiplexing Attack*: For instance, Fig. 2(a) involves a malicious SMF that randomly triggers various procedures toward the AMF, adhering to HTTP/2 protocol precedence constraints [6], [43]. For *Attack 1.1-SMA-Request/Response*, we use three different procedures, that are triggered from the malicious SMF towards the AMF using the same `Namf_Communication_N1N2MessageTransfer` API, such as Uplink, downlink, and UE release PDU session. Note that this API covers most of the service operations provided by the AMF and consumed by the SMF [43]. As attackers, we repeat this attack over 55,954 HTTP/2 connections, each handling up to 907 requests, resulting in an NFp overload and a DoS. The second attack scenario (Fig. 2(b)) considers an SMA that involves a malicious SMF exploiting 30 UEs by triggering only the Downlink procedure using `Namf_Communication_N1N2MessageTransfer` API, however, we include a notify URI for DISCONNECTED UEs. According to 3GPP specifications [43], when the Downlink procedure is initiated while the UE state is DISCONNECTED, the `N1N2TransferFailureNotification` API is triggered to notify SMF that the UE is unreachable [43]. Consequently, the AMF sends a notification back to the malicious SMF. We emulated this attack using 54,188 HTTP/2 connections, each handling up to 841 requests over 40 minutes before the AMF goes down. This attack effectively exploits the signaling mechanisms of the network, leading to a DoS on the AMF and SMF. The continuous failure notifications overwhelm both AMF and SMF, making it unresponsive and crippling NFs, degrading the Quality of Service (QoS) for legitimate UEs.

To better illustrate how we perform *Attack 1.1-SMA-Request/Response*, we illustrate in Fig. 5 the normal Downlink procedure triggered from the Data Network (DN) when the UE is in the IDLE state [43], and highlight in red how an attacker can perform the attack as in our emulation assuming that the SMF was compromised. In a normal scenario, the SMF sends a request to the AMF using `Namf_Communication_N1N2MessageTransfer` API (Fig. 5(3a)). The AMF responds to the SMF indicating that the UE is not reachable, and subsequently sends a Paging Request to the UE/RAN (Fig. 5(4b)). The Paging Request triggers the Uplink procedure to activate the UE. In the dashed red box (Fig. 5), we highlight a scenario where an attacker compromises the SMF and launches a malicious `Namf_Communication_N1N2MessageTransfer` request towards the AMF, triggering a Downlink procedure for a UE in IDLE

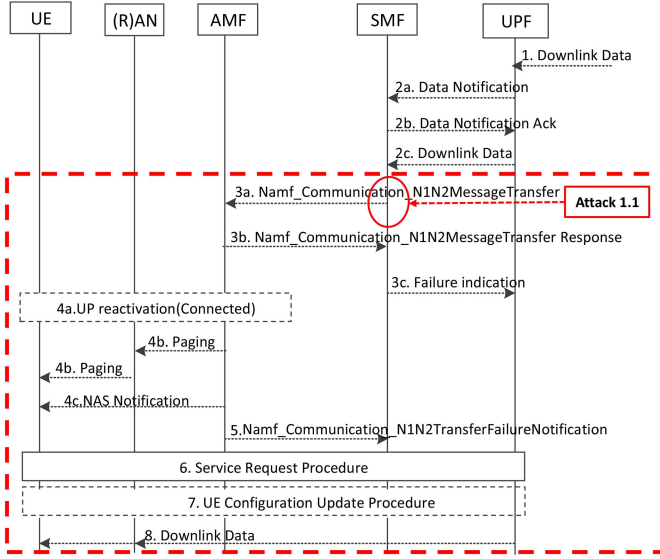


Fig. 5. Network triggered service request procedure [43].

state without requiring network signaling. Although the attacker only initiates a single request, it results in a chain of other messages (Fig. 5(3b-8)) related to paging, service request, PDU session update, and PDU session modification in the 5G network. This depicts a high overhead that an attacker can introduce to the network with a single malicious request.

2) *Attack 2: HTTP/2 Rapid Reset Attack*: In our emulation of *Attack 2-Rapid Reset*, a compromised PCF targets the UDR using a UDR Management procedure [42]. We assume that the UDR sets its HTTP/2 connection with *SETTINGS_MAX_CONCURRENT_STREAMS* = 1000. As depicted in Fig. 2(c), the PCF sends to the UDR a request using *Nudr_DataManagement* API followed by a *RST_STREAM* frame to stop the sent request. The malicious PCF establishes around 263,251 HTTP/2 connection with the UDR over 2 hours, with up to 2306 requests and *RST_STREAM* frame per connection. This action aims to create a DoS situation, effectively disrupting the network's operations and impacting its ability to process legitimate requests.

3) *Attack 3: HTTP/2 Slow Rate Attack*: We emulate three variations of HTTP/2 slow rate attack from PCF to UDR. As depicted in Fig. 2(d), to emulate the *Attack 3.1-Slow Rate-Setting*, the PCF establishes around 3,947 HTTP/2 connections with the UDR and sends to it on each of them a *SETTINGS* frame. As the UDR is malicious, it does not acknowledge the *SETTINGS* frames sent by the PCF, leading to a backlog of unacknowledged frames, hence, causing a drop of these connections after a certain timeout time. We emulate the *Attack 3.2-Slow Rate-Connection Preface* (Fig. 2(e)) by accounting for a malicious PCF that sends a connection preface to the UDR without following it by any HTTP/2 GET/POST. This makes the UDR wait endlessly for an HTTP/2 request that never arrives. This scenario is repeated over 5,733 HTTP/2 connections and exhibits similar behavior to normal network traffic. During *Attack 3.3-Slow Rate-Window Size* (Fig. 2(f)), the malicious PCF establishes the first HTTP/2 connection and sends a manipulated HTTP/2 *SETTINGS* with

SETTINGS_INITIAL_WINDOW_SIZE equal to zero, signaling that the PCF can no longer receive data. This manipulation forces the UDR to halt all data transmissions until it receives a *WINDOW_UPDATE* frame, effectively freezing the data flow. The malicious PCF repeats this attack over 3,815 HTTP/2 connections severely impacting the 5G network availability.

VI. ATTACKS IMPACT AND PREVENTION

A. HTTP/2 Attacks Impact

Our emulations of HTTP/2 attacks on 5G SBA presents various impact on NFs resource utilization that we measure through observing the CPU consumption of the different NFs in Fig. 6. Additionally, we measure the total number of messages (i.e., requests and notifications only) exchanged per each pair of NFs within the 5G SBA during 20 minutes of the different attacks as shown in Fig. 7. This metric reflects the volume of control signaling traffic impacted by the HTTP/2 attacks especially when compared to the benign traffic captured for the same period during the benign emulation (Fig. 4). We analyze in the following the impact of the different attacks on 5G networks, suggest prevention and mitigation measures, and summarize them in Table II.

1) *Attack 1: HTTP/2 Stream Multiplexing Attack*: Upon the start of *Attack 1.1-SMA-Request/Response* (Fig. 6(a)) and *Attack 1.2-SMA-Subscribe/Notify* (Fig. 6(b)) at time 60 (i.e., after around an hour of emulations), the CPU usage of the AMF and SMF increases sharply, while the CPU usage for the rest of the NFs decreases. This is mainly attributed to the overload at the AMF and SMF, delaying and potentially blocking the completion of 5G procedures that are stuck at the SMF-AMF interactions. As shown in Fig. 6(a), at time 104 (i.e., after 44 minutes of the start of the attack), the AMF fails, resulting in a DoS. Further, the regular CPU spikes during *Attack 1.2-SMA-Subscribe/Notify* (Fig. 6(b)) highlight intense activity periods that stress the SMF and potentially degrade services of other NFs. Notably, we observe the highest number of messages during SMF-AMF interaction due to frequent attack requests from the SMF towards the AMF as illustrated in Fig. 7(a) and (b). Additionally, the number of messages in the AMF-SMF interactions during *Attack 1.2-SMA-Subscribe/Notify* (Fig. 7(b)) is higher than during the *Attack 1.1-SMA-Request/Response* (Fig. 6(a)) due to the notifications sent from the AMF to the SMF. In summary, SMA attacks significantly impact network availability by exhausting its resources, and causing a DoS on the targeted NF and potentially on the whole 5G network.

2) *Attack 2: HTTP/2 Rapid Reset Attack*: Although the *Attack 2: HTTP/2 Rapid Reset* was emulated while the official patch was deployed in our network, we notice that the increase in the *SETTINGS_MAX_CONCURRENT_STREAMS* (Section IV-C) can still disrupt the provided QoS, not only by overloading the attack target (i.e., UDR) but also the attack source (i.e., PCF). Fig. 6(c) shows high CPU consumption at the UDR and PCF that often reaches 80% for an hour. However, after 2 hours of running the attack, we observe CPU spikes reaching 160% between times 120 and 180, indicating moments of intense load on the UDR, always accompanied by a high load

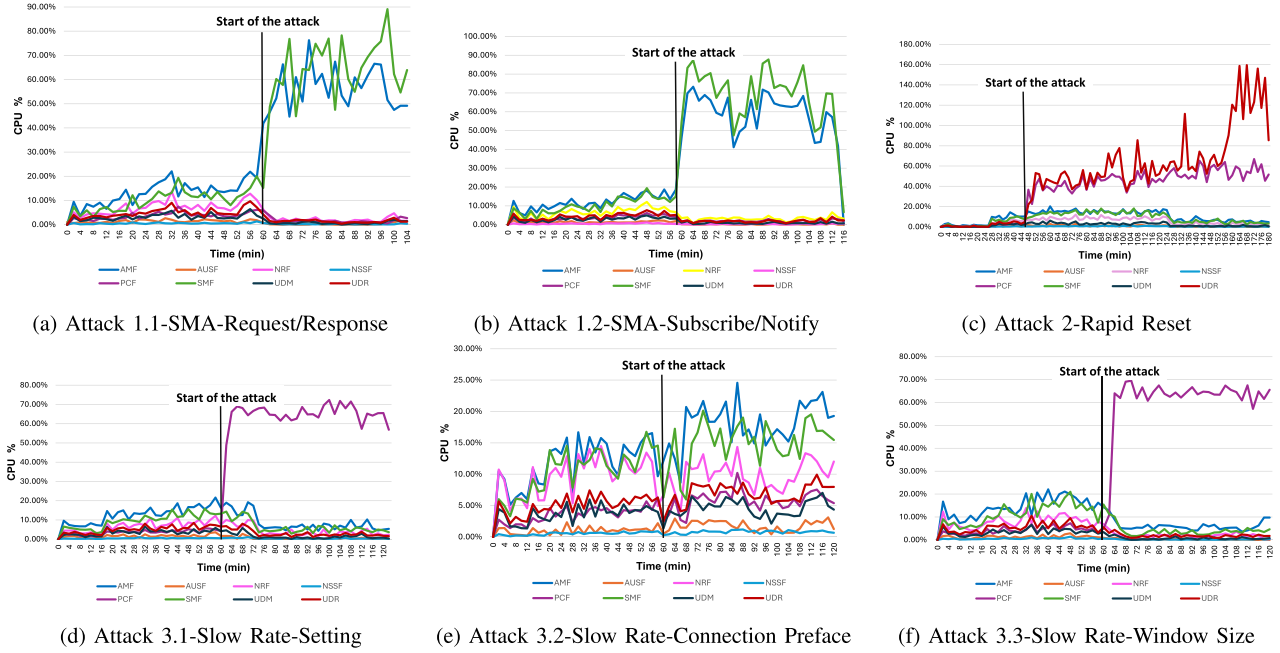


Fig. 6. 5G SBA NFs CPU consumption during malicious network behavior.

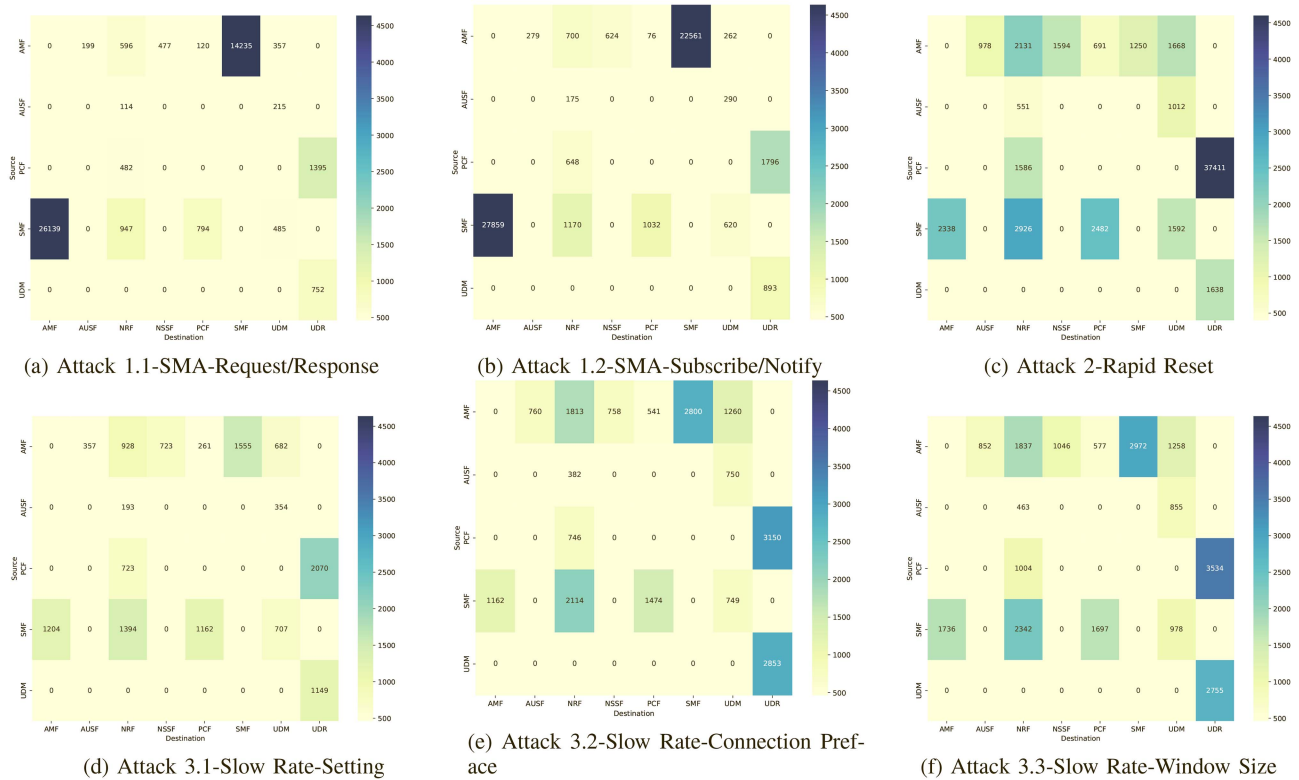


Fig. 7. Total number of messages between pairs of NFs in 5G SBA during malicious network behavior.

on the PCF. This is also reflected by the high number of messages exchanged between PCF and UDR in Fig. 7(c). However, a lower CPU consumption is observed at the remainder NFs between times 120 and 180 reflecting a DoS attack on the network and a degradation of the QoS of those NFs (Fig. 6(c)).

3) *Attack 3: HTTP/2 Slow Rate Attack:* When examining the variations of HTTP/2 slow rate attacks, we notice their distinct impacts on the NFs within the 5G SBA, starting at attack time 60. Fig. 6(d) and (f) reflect a high CPU consumption on the targeted NF, the PCF, with a degradation of the CPU consumption on the

TABLE II
HTTP/2 ATTACKS, IMPACT AND PROTECTION MEASURES IN 5G SBA

Attack Type	Description	Impact on the network	Protection measures
Attack 1.1-SMA-Request/Response	SMF -> AMF: A malicious SMF initiates multiple 5G requests towards an AMF by exploiting the stream multiplexing feature over many HTTP/2 connections.	Overload of attack NFs (i.e., SMF, AMF), complete DoS on the network.	Anomaly detection systems, limiting <code>SETTINGS_MAX_CONCURRENT_STREAMS</code> on NFp.
Attack 1.2-SMA-Subscribe/Notify	SMF -> AMF: A malicious SMF exploits the stream multiplexing feature by triggering the Downlink procedure for DISCONNECTED UEs towards the AMF with a notification subscription. AMF then informs the SMF of UE's unreachability using the N1N2TransferFailureNotification API.	Overload of attack NFs (i.e., SMF, AMF), complete DoS on the network.	Intelligent adjustment of <code>SETTINGS_MAX_CONCURRENT_STREAMS</code> value, anomaly detection systems.
Attack 2-Rapid Reset	PCF -> UDR: A malicious PCF initiates the UDR Management procedure towards the UDR to trigger a registration procedure and immediately sends a <code>RST_STREAM</code> frame.	Overload of attack NFs (i.e., PCF, UDR), degradation of other NFs QoS.	Limiting the number of <code>RST_STREAM</code> frames received on NFp, Anomaly detection systems.
Attack 3.1-Slow Rate-Setting	PCF -> UDR: The PCF sends multiple requests to the malicious UDR, starting with a <code>SETTING</code> frame, but the malicious UDR does not acknowledge any received messages.	Resource exhaustion at targeted NF, degradation of other NFs QoS.	Anomaly detection systems, intelligent monitoring systems, and timer values to drop or close malicious connections.
Attack 3.2-Slow Rate-Connection Preface	PCF -> UDR: A malicious PCF sends a connection preface to the UDR but never sends any HTTP/2 requests, causing the UDR to wait until its connection pool is full, eventually dropping the connection.	Similarities to normal network traffic, a longer period of unnoticed resource depletion impact.	Anomaly detection systems.
Attack 3.3-Slow Rate-Window Size	PCF -> UDR: A malicious PCF sends an HTTP/2 <code>SETTINGS</code> frame with <code>SETTINGS_INITIAL_WINDOW_SIZE=0</code> to the UDR, halting all data transmission. The UDR waits indefinitely for a <code>WINDOW_UPDATE</code> frame, which the malicious PCF intentionally never sends.	Resource and connection pool exhaustion at targeted NF, degradation of other NFs QoS.	Anomaly detection systems, intelligent monitoring systems and timer values to drop or close malicious connections.

other NFs, reflecting a degradation of the network availability and QoS without observing a total DoS. Although the PCF CPU consumption is comparable in both, *Attack 3.1-Slow Rate-Setting* and *Attack 3.3-Slow Rate-Window Size*, we notice from Fig. 7(d) and (f) that the number of messages exchanged between UDR and PCF is higher in *Attack 3.3-Slow Rate-Window Size* than in *Attack 3.1-Slow Rate-Setting*. This explains that the high CPU consumption is not a result of the number of exchanged messages but rather of resources allocated during the waiting times at the PCF for a `SETTINGS` acknowledgment in case of *Attack 3.1-Slow Rate-Setting* and for a `WINDOW_UPDATE` in case of *Attack 3.3-Slow Rate-Window Size*.

Attack 3.2-Slow Rate-Connection Preface does not exhibit a significant impact on the CPU of the different NFs, as illustrated in Fig. 6(e). In contrast, although the trend is different, the CPU consumption is comparable to the benign network behavior shown in Fig. 4(b). This indicates that detecting this attack may be more challenging. By observing Fig. 7(d), (e), and (f), we note that the total number of messages exchanged during the different HTTP/2 slow rate attacks is lower than that exchanged during normal network operations (Fig. 4(b)). This demonstrates that simply counting messages between NFs is insufficient to detect manipulation in the HTTP/2 flow, particularly in HTTP/2 frames.

B. Discussion and Mitigation Measures

The analysis of HTTP/2 attacks on our 5G testbed reveal that HTTP/2 SMA attacks are the most damaging due to the DoS impact they cause on the targeted NF and on the 5G network as a whole. More specifically, degradation of the performance of the 5G NFs not directly involved in the attack is observed and is highly related to the type of the targeted NFs. For instance, the emulated 5G procedures (Table I) involve many interactions between the AMF and SMF that cause a bottleneck for the

completion of these procedures which was reflected by a degradation of the CPU consumption of other NFs which were not receiving as many messages as during the normal network traffic emulations. Similar performance degradation was also observed on the NFs not involved in the attack during the rapid reset and slow rate attacks. However, these attacks exploit the inherent limitations in timeout and rate-handling mechanisms within the network, pushing the 5G SBA towards a slow degradation rather than a sudden failure. This subtlety can lead to longer periods of unnoticed impact, allowing significant damage over time.

It is worth noting that the value assigned to the `SETTINGS_MAX_CONCURRENT_STREAMS` plays an important role in protecting the network against SMA and rapid reset attacks. Here, an intelligent solution for setting the value of this `SETTINGS_MAX_CONCURRENT_STREAMS` is highly important to achieve the trade-off between network security and network performance. A high value of this setting can maximize the benefits of the stream multiplexing feature in terms of latency, however, it can increase the SMA impact on the network. Thus, an intelligent and adaptive `SETTINGS_MAX_CONCURRENT_STREAMS` value adjustment solution based on network state can be efficient in protecting the network against SMA attacks. The stealthy nature of slow-rate attacks can make their detection challenging, requiring intelligent anomaly detection solutions. In contrast, they can be prevented by intelligent monitoring solutions and timer values to drop or close malicious connections with long inactivity time at the HTTP/2 client. Although these attacks are exploited in the web, exploiting these attacks in a 5G environment requires attacking 5G specific API, making these attacks more challenging to perform than in the web where APIs are not necessarily used. Furthermore, their impact on 5G network can be more disruptive than in a web environment given the dependencies and interactions existing between 5G NFs.

TABLE III
SELECTED FLOW-BASED FEATURES [50]

Selected Flow-based Features
Src Port, Dst Port, Protocol, TotLen Fwd Pkts, TotLen Bwd Pkts, Fwd Pkt Len Max, Fwd Pkt Len Min, Fwd Pkt Len Mean, Fwd Pkt Len Std, Bwd Pkt Len Max, Bwd Pkt Len Min, Bwd Pkt Len Mean, Bwd Pkt Len Std, Flow Byts/s, Bwd IAT Std, Fwd PSH Flags, Bwd PSH Flags, Fwd URG Flags, Bwd URG Flags, Fwd Header Len, Bwd Header Len, Pkt Len Min, Pkt Len Max, Pkt Len Mean, Pkt Len Std, Pkt Len Var, FIN Flag Cnt, SYN Flag Cnt, RST Flag Cnt, PSH Flag Cnt, ACK Flag Cnt, URG Flag Cnt, CWE Flag Count, ECE Flag Cnt, Pkt Size Avg, Fwd Seg Size Avg, Bwd Seg Size Avg, Fwd Byts/b Avg, Fwd Pkts/b Avg, Fwd Blk Rate Avg, Bwd Byts/b Avg, Bwd Pkts/b Avg, Bwd Blk Rate Avg, Subflow Fwd Byts, Subflow Bwd Pkts, Subflow Bwd Byts, Init Fwd Win Byts, Init Bwd Win Byts, Fwd Act Data Pkts, Fwd Seg Size Min, Active Mean, Active Std, Active Max, Active Min

VII. DATASETS

To generate a 5G dataset that mirrors real 5G network traffic, we emulate normal and attack traffic as noted in Section V and collect the generated data. Using Wireshark [49], a network monitoring tool, we capture network traffic data within our 5G testbed. These captures are raw packets stored as PCAP files [49], and document raw network interactions between various entities like UE, RAN, and 5G NFs. We collect benign data and execute HTTP/2 attacks at different times. We refine our dataset by processing raw network-layer data with CICFlowMeter [50], which generates 84 flow-based features capable of distinguishing normal and malicious behaviors [29], [30], [31]. Each row in the resulting CSV file represents a single flow, defined as packets with the same source IP, destination IP, source port, and destination port within a specified time interval. Separate CSV files were created for benign traffic and each emulated HTTP/2 attack.¹

Following the extracted features, we perform feature normalization and select the most relevant ones. At the feature selection stage, we use the variance threshold [51] function to determine the most relevant variance value of the features. We choose this selection function, as it is well known for its usage in unsupervised models [51]. The purpose of its usage is to help in removing features with minimal variations or those deemed as noise. As the model is highly dependent on 5G SBA behavior patterns, the features selected to train the model must be accurately represented (i.e., have high variance) and provided to the anomaly detection module, as a result, we consider 54 features that have high variance, as shown in Table III.

For each emulated scenario, we divide the flow-based dataset (Table IV) into two categories: benign and attack, with the total duration of each emulation. Although emulations were planned for two hours, *Attack 1.1-SMA-Request/Response* and *Attack 1.2-SMA-Subscribe/Notify* lasted for 1 hour 40 minutes and 1 hour 50 minutes, respectively, as the network went down due to the attack. It is worth noting that the reported datasets in Table IV

TABLE IV
FLOW-BASED DATASET IN 5G NETWORKS

Emulation Type	Benign Rows	Attack Rows	Duration
Benign	129 367	-	2 hours
A.1.1-SMA-Req/Resp	90 163	55 954	1 hour 40 minutes
A.1.2-SMA-Sub/Not	91 010	54 188	1 hour 50 minutes
A.2-Rapid-Reset	135 120	314 866	3 hours
A.3.1-SR-Setting	74 540	11 696	2 hours
A.3.2-SR-Con-Pref	53 722	10 278	2 hours
A.3.3-SR-Wind-Size	68 134	16 738	2 hours

are mutually exclusive and do not include any redundant records. For the data to be usable for anomaly detection, we label our flows as benign (0) and attack (1) based on our knowledge of the compromised UEs used and the time of the attack emulations were launched.

VIII. HTTP/2 ANOMALY DETECTION

In this section, we evaluate the performance of three unsupervised models using flow-based features as an anomaly detection solution in 5G SBA, focusing on their ability to identify HTTP/2-5G-specific attacks.

A. Anomaly Detection Benchmark Models

To detect HTTP/2 attacks in 5G networks, we adopt unsupervised machine learning models due to the lack of labeled real-world data and the need to detect zero-day attacks [52]. In 5G SBA, compromised Network Functions (NFs) may act maliciously while still appearing legitimate, and the origin of such attacks is often unknown. Unsupervised models are therefore well-suited to this threat landscape.

Among the unsupervised models, we first include the AE. AEs are designed to learn compact representations of normal data through reconstruction, such that large reconstruction errors indicate deviations or anomalies [53]. We selected the AE model not only because of its proven applicability in general anomaly detection, but also to evaluate its performance beyond its original design context in [13], where we previously developed an AE-based solution specifically for detecting HTTP/2 SMA. In this work, we challenge the AE's ability to generalize by testing whether a model effective against a single known attack (SMA) can also identify five additional HTTP/2 attacks that exhibit different traffic characteristics and complexities. This allows us to evaluate the generalization capability of AE for broader 5G HTTP/2 attack detection.

We further include the LSTM-based Autoencoder (LSTM-AE) to account for temporal patterns in 5G network traffic [54]. 5G SBA communication flows exhibit sequential dependencies due to service chaining, stateful procedures, and inter-NF signaling. LSTM layers can capture these dependencies, making LSTM-AE particularly effective for recognizing complex, multi-stage attacks that develop over time. Our choice is also motivated by the work in [19], which successfully employed LSTM-AE for detecting non-HTTP/2-based attacks in 5G SBA. By adopting and adapting the same model in this study, we extend its application to new attack classes specific to HTTP/2, assessing its robustness across a wider threat spectrum in the 5G context.

¹The different CSV files from our dataset will be made publicly available upon the paper acceptance.

TABLE V
HYPERPARAMETERS

Hyperparameter	Architecture	Dropout	Batch size	Optimizer	Hidden activation	Estimator
LSTM-Autoencoder	[32,16,16,32]	0.1	16	Adam	ReLU	-
Autoencoder	[42,2,42]	-	32	Adam	ReLU	-
Isolation Forest	-	-	-	-	-	50

TABLE VI
F1-SCORE OF LSTM-AE, AE, IF ACROSS HTTP/2 ATTACKS

Attack Type	LSTM-Autoencoder	Autoencoder	Isolation Forest
A.1.1	93.78%	80.16%	82.63%
A.1.2	97.58%	86.08%	88.42%
A.2	96.08%	81.73%	82.32%
A.3.1	88.21%	84.50%	83.22%
A.3.2	89.45%	85.09%	85.23%
A.3.3	88.32%	82.11%	84.22%
Average	92.24%	83.28%	84.34%

Finally, we include the Isolation Forest (IF) as a lightweight, scalable, and interpretable unsupervised model. IF is effective for detecting outliers in high-dimensional data by recursively partitioning data points based on randomly selected features. Its core strength lies in its independence from distribution assumptions and its capability to isolate rare anomalies quickly [55]. This is crucial in our use case, where the attacker may compromise any NF (NFc or NFp) and initiate attacks without triggering obvious symptoms. Since we do not assume prior knowledge of the attack's origin, and the dataset is largely unlabelled, IF allows us to detect anomalous flows without explicit supervision, relying instead on the statistical rarity of malicious behavior in the flow-based feature space. Thus, we train and compare the performance of LSTM-AE, AE, and IF in detecting HTTP/2 attacks.

B. Experimental Results

We evaluate the performance of the three aforementioned models in detecting the six HTTP/2 attacks that we emulated. We use the F1-score, an effective evaluation metric to assess the models' precision and recall capabilities and hence, their detection performance. We first train and validate multiple architectures for each of the three models, selecting the one with the best detection performance. The final chosen architecture and hyperparameters for each model are reported in Table V. To accomplish this, we allocate 20% of the training dataset of size 100,000 rows as a validation dataset, and we train the models using the remaining training dataset. To test the three models and to check their performance over different HTTP/2 attacks, we select from each attack file (Table IV) 30,000 benign rows and 10,000 attack rows.

After training and validating the models, we test each model using the test dataset. As shown in Table VI, LSTM-AE outperforms AE and IF across all HTTP/2 attacks. LSTM-AE achieves an average F1-score of 92.24% across HTTP/2 attacks, reflecting its robustness in anomaly detection. IF follows with a lower F1-score. AE shows a comparable average F1-score to IF, but a lower F1-score particularly in more complex attack scenarios (i.e., attack 1.1, attack 3.3), indicating its relative difficulty in capturing all anomalies compared to other models. Fig. 8 shows a

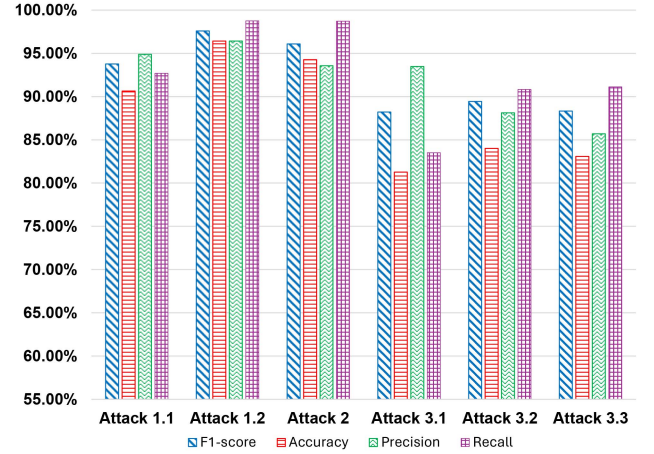


Fig. 8. LSTM-AE performance across HTTP/2 attacks.

detailed performance of LSTM-AE which consistently achieved the highest F1-score across all HTTP/2 attacks, highlighting its superior ability to capture temporal dependencies in 5G network data. Notably, it has higher precision for *Attack 1.1* or *Attack 3.1* depicting the ability of LSTM-AE to detect them. However, a higher recall is obtained for the remaining attacks, showing the model struggles to correctly detect them. The results suggest that although LSTM-AE is the most robust, further fine-tuning and optimization of all models are necessary to enhance their performance, especially in complex scenarios.

To better evaluate the LSTM-AE model across six distinct attacks, we rely on the Receiver Operating Characteristic (ROC) curves. An ROC curve illustrates the trade-off between the False Positive Rate (FPR) and the True Positive Rate (TPR) across all thresholds [56]. The Area Under the ROC Curve (AUC) is a commonly used metric in conjunction with the ROC curve, providing an aggregated measure of the model's performance over all thresholds. An $AUC = 1$ indicates a perfect model capable of achieving $TPR = 1$ and $FPR = 0$ with an ideal threshold. Fig. 9 showcases the performance of the LSTM-AE model tested over six attacks. With AUC values ranging from 0.87 to 0.97 across HTTP/2 attacks, the results highlight the model's ability to detect anomalies effectively. However, the variation in AUC across attacks emphasizes the impact of attack complexity and feature relevance on the detection performance.

We analyzed the LSTM-AE model's training and detection time efficiency to examine its scalability. The model processes input batches with low memory overhead and <50 millisecond detection time per 1,000 records on a standard GPU, making it suitable for near real-time detection tasks. Its architecture also supports parallelization across slices or functions. These characteristics suggest that the model can scale to larger deployments by replicating across multiple nodes in a 5G environment, maintaining consistent performance.

IX. COMPARISON WITH THE STATE OF THE ART

Despite the increasing reliance on HTTP/2 as the primary signaling protocol in 5G SBA, there remains a significant gap in

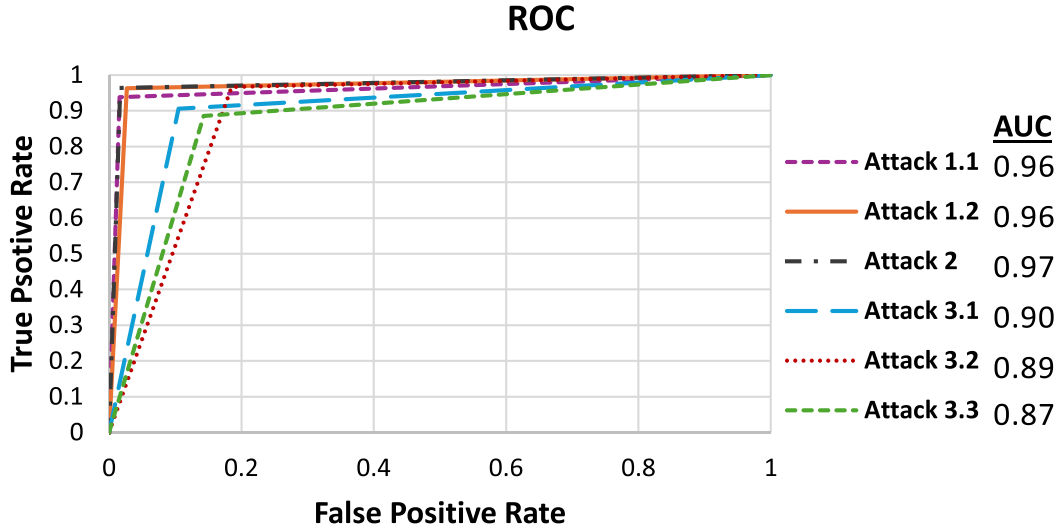


Fig. 9. AUC-ROC of LSTM-AE across HTTP/2 attacks.

the literature regarding the empirical analysis and detection of HTTP/2-based attacks within this context. Existing works either lack experiments or provide only a limited evaluation of specific attack types. For example, [11], [12] provide general overviews of signaling security in 5G networks and mention potential HTTP/2-related risks. However, their studies are theoretical and do not include simulation, detection, or attack modeling. Similarly, [57] highlighted flaws in HTTP/2 design but focused on general web environments rather than 5G-specific systems.

Several studies have experimentally analyzed HTTP/2 attacks, but mostly outside the 5G context. [25], [26], [58] demonstrated slow-rate DoS attacks using settings flooding, connection preface abuse, and delayed streams, proposing detection based on event-sequence analysis. [22], [28] explored asymmetric multiplexing-based attacks, while [59] proposed intelligent feature selection techniques for detecting HTTP/2 threats. However, these efforts did not consider the 5G SBA architecture, limiting their relevance to 5G networks. In addition, [14] focused on HTTP/2 attack traffic generation using 5Greplay, allowing replay-based simulation of protocol-level threats. While the tool supports traffic generation, the study did not include evaluations of detection performance, impact on 5G functions, or dataset release for benchmarking.

In contrast, [13] is the only work that emulates HTTP/2 attacks in 5G SBA. While this study takes an important step forward, it focuses exclusively on SMAs and lacks coverage of other critical HTTP/2 attack vectors, such as rapid reset and slow-rate attacks. Furthermore, [13] does not release its dataset publicly, limiting reproducibility and comparative research across the field. The detection method used in [13] relies on AE. We test the AE with the six HTTP/2 attacks, which report an average F1-score of 83.28%. These results surpass those reported in [13] in terms of detection accuracy and the range and realism of attack scenarios covered.

Our work addresses these limitations by significantly expanding the attack coverage and introducing a detailed,

practical evaluation of six HTTP/2 attack variants: two SMAs (Request/Response and Subscribe/Notify), one rapid reset attack (CVE-2023-44487), and three slow-rate attacks (based on settings frame, connection preface, and window size manipulation). All attacks were implemented and evaluated on a fully standalone 5G testbed, adhering to 3GPP standards to ensure a realistic emulation of 5G network interactions and control plane behavior. Additionally, this study is the first to generate and publish a comprehensive dataset capturing both benign and malicious HTTP/2 traffic within a real 5G SBA environment. This dataset includes flow-based features commonly used in anomaly detection and is structured to support reproducibility, benchmarking, and further research on 5G network security.

In summary, unlike [11], [12], which remains theoretical, and [13], which is limited in scope and effectiveness, our work delivers a comprehensive, practical, and publicly reproducible framework for evaluating the impact of HTTP/2 attacks on 5G SBA. It also provides a solid benchmark for developing and testing advanced anomaly detection models specifically tailored to the security challenges of HTTP/2 in next-generation mobile networks.

X. CONCLUSION

In conclusion, this work presented the first practical study of various HTTP/2 attacks in 5G SBA using the open-source free5GC testbed and UERANSIM, adhering to 3GPP standards. For instance, our emulation showed that an attacker who managed to compromise an NF can cause harm to a 5G network through HTTP/2 DoS attacks and that HTTP/2 settings configuration across the different 5G NFs play a crucial role in securing 5G networks against such exploitation. Further, our emulations were leveraged to offer the first public dataset for anomaly detection in a 5G network. The dataset depicted flow-based features of benign and six different HTTP/2 attacks traffic, including two versions of HTTP/2 stream multiplexing attacks, HTTP/2 rapid

reset attacks, and three variations of HTTP/2 slow rate attacks. Finally, we presented three machine learning models, including AE, LSTM-AE, and IF, that detected HTTP/2 attacks with a respective average F1-score of 92.24%, 83.28%, and 84.34%. These models can serve as benchmarking algorithms for future research.

Future work will focus on refining and combining the proposed models through ensemble learning, while incorporating diverse feature types such as flow-based, temporal, and protocol-specific features to improve detection accuracy and robustness against sophisticated and evolving HTTP/2 threats in 5G networks.

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